

CENTER FOR MEDICAL INNOVATION





EOSINOPHILIC ESOPHAGITIS TESTING:

NOVEL EOE BIOMARKER SAMPLING DEVICE(S)

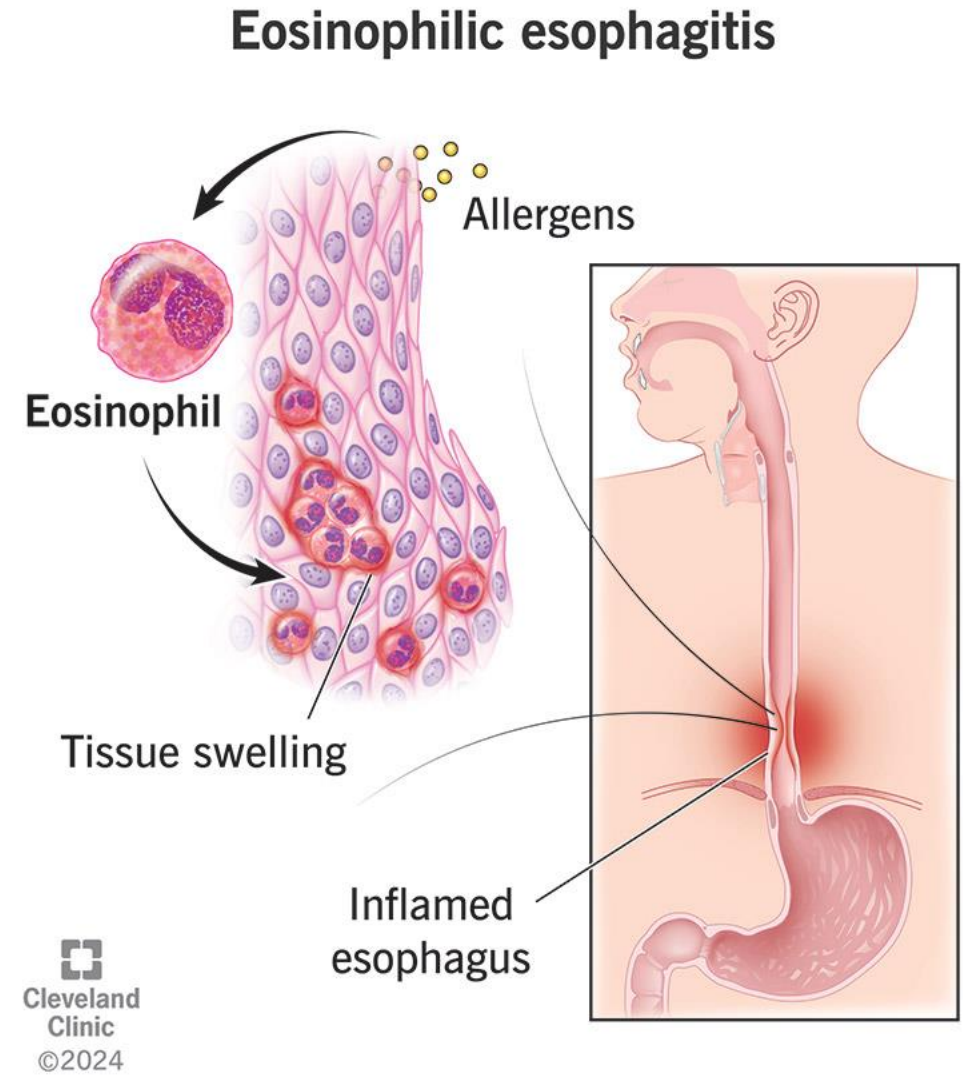
DR. KATHRYN PETERSON – PRINCIPAL INVESTIGATOR
DILLON DESPAIN – PROJECT ENGINEER
AMANDA LEMATTY, YASH AGRAWAL – MARKET ASSESSMENT

OVERVIEW

EoE is an allergic condition that happens in the esophagus. The esophagus becomes inflamed and does not contract properly.

The most common symptom is dysphagia, or trouble swallowing.

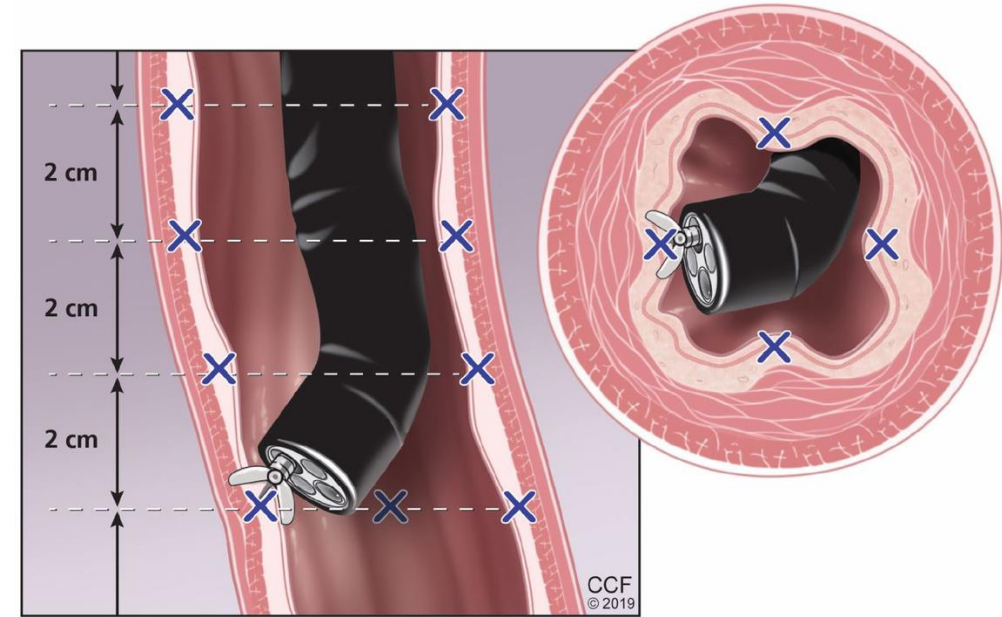
EoE affects 42,000 people in the US annually.



OVERVIEW - Continued

Current EoE detection procedure:

- Sedation of patient.
- Collection of biopsies (4-8) via an endoscope using forceps (see image right).
- Analysis by pathology to count eosinophils in the samples.
- Patients require an additional 1-8 endoscopies depending on treatment plan.



CHALLENGES

1

Cost/Difficulty of Repeat Scoping

Difficult to track the disease state with current procedure. Sedation during upper endoscopies is difficult and expensive on patients.

2

Accuracy and Time for Pathology Results

Need to wait up to two weeks on pathology. Pathology count has accuracy concerns in correlation to EoE.

INTERVIEWS AND OBSERVATIONS



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Judith Staub, MD
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“Endoscopy is a very involved and expensive way to follow the disease. A way to quantify EoE more simply would be very useful.”

NOVEL BIOMARKER CORRELATIONS

Dr. Peterson has published peer reviewed research on correlation of other biomarkers to EoE:

- IgG4
- eMBP
- Tryptase
- These are tested with spectroscopy or a tryptase test that can be tested in a lab similar to ARUP.

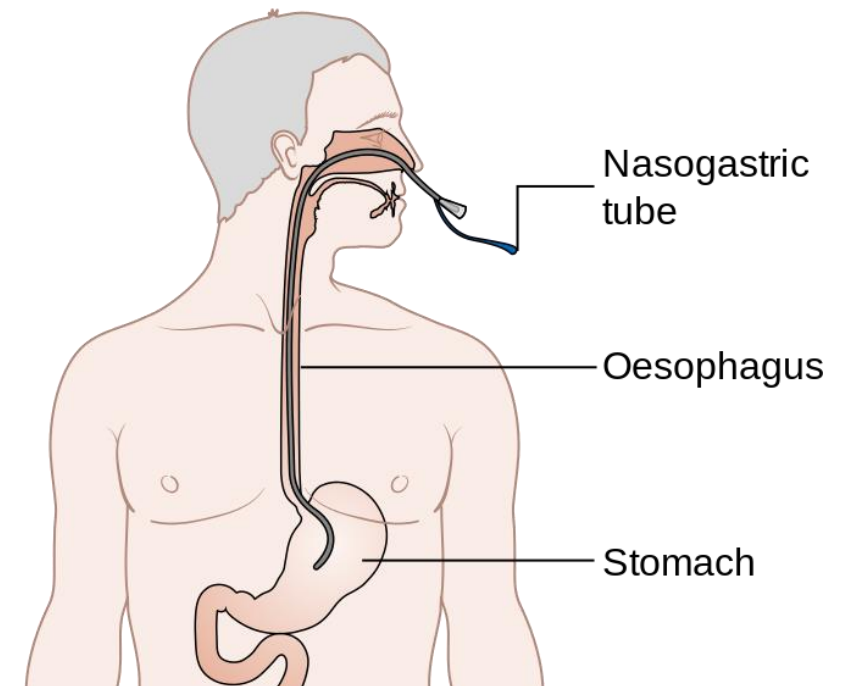
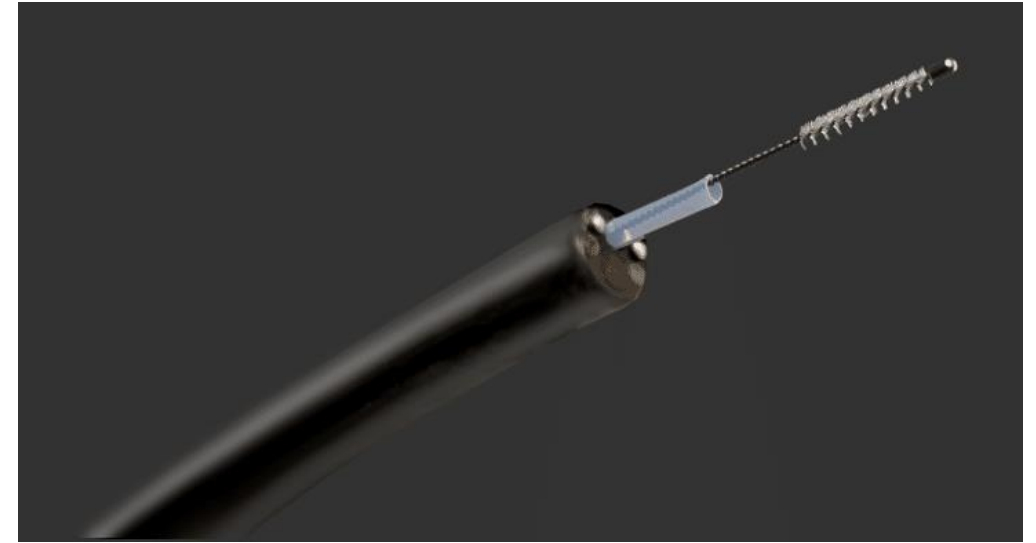
PROPOSED DEVICE(S)

Device #1 – Endoscope mucus sampler

- The first device needed is similar to the cytology brush pictured (for cell sampling).
- Diameter sized to fit through endoscope
- Have an absorbent sponge or other collection method, i.e. suction design to maximize mucus volume.
- Easily store/transfer mucus sample.

Device #2 – Nasal mucus sampler

- This device would be similar to a nasogastric tube.
- Introduced trans nasally as a standalone device.
- Collects mucus vis suction/absorption and then removed nasally.
- Allows for collection without sedation/endoscopy.
- Potential for



MARKET ASSESSMENT

Total Addressable Market:

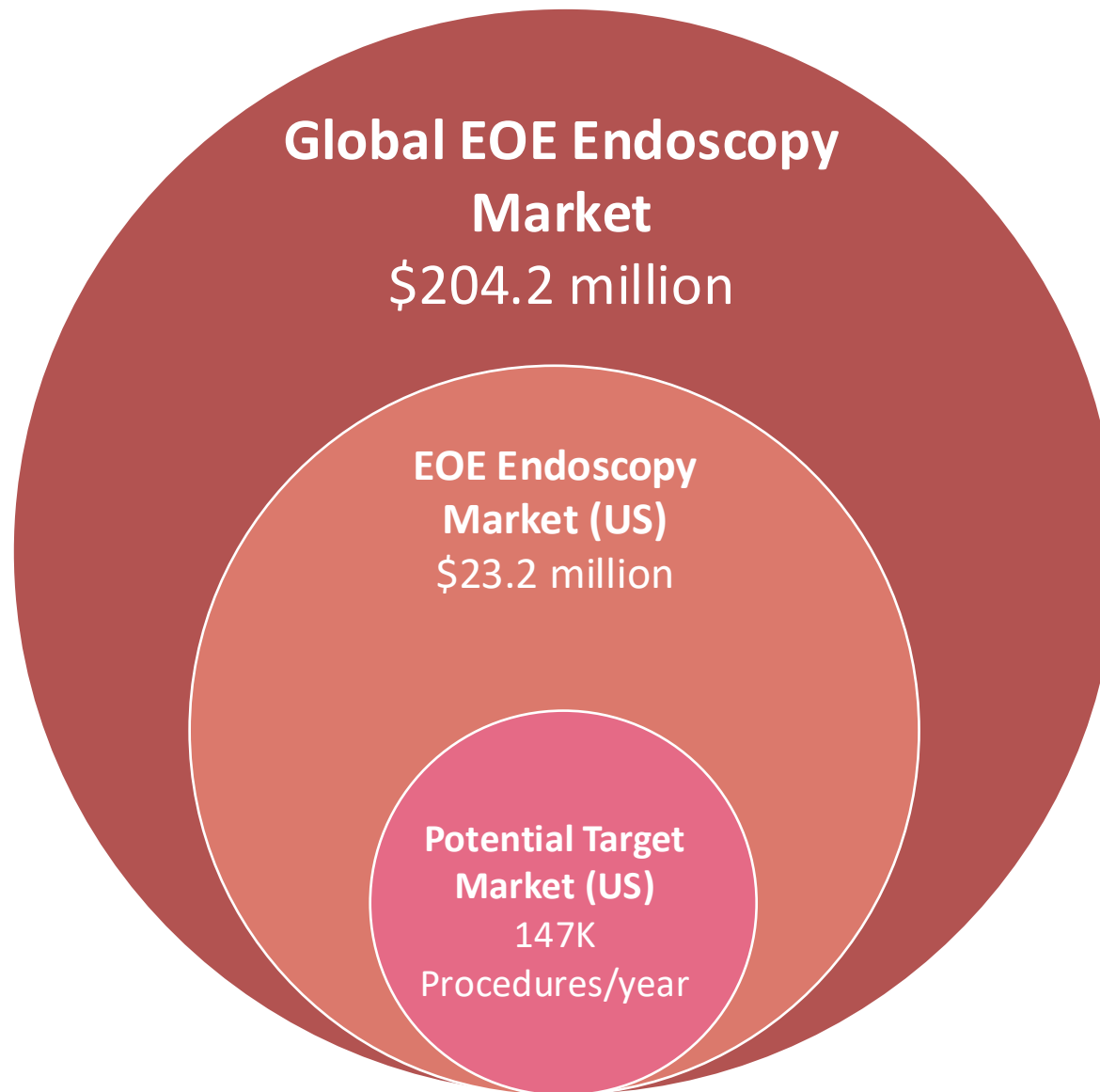
Global EOE Endoscopy Market: **\$204.2 million**

Serviceable Available Market:

US EOE Endoscopy Market: **\$23.2 million**

Potential Target Markets:

- **Worldwide : 1.3 million** procedures/year
- **US : 147 thousand** procedures/year



POTENTIAL COMPETITION

Skin Prick/Food Patch Tests

- YorkTest Laboratories
- \$60-\$500
- Tests for IgE on the surface of skin
- 50-60% false positives
- There is concerns with IgE's correlation to EoE

P88 Dietary Antigen Tests

- Precision Point Diagnostics
- Tests for a variety of food allergies/sensitives using IgE, IgG4, IgG, C3d
- \$500-600

REIMBURSEMENT CONSIDERATIONS

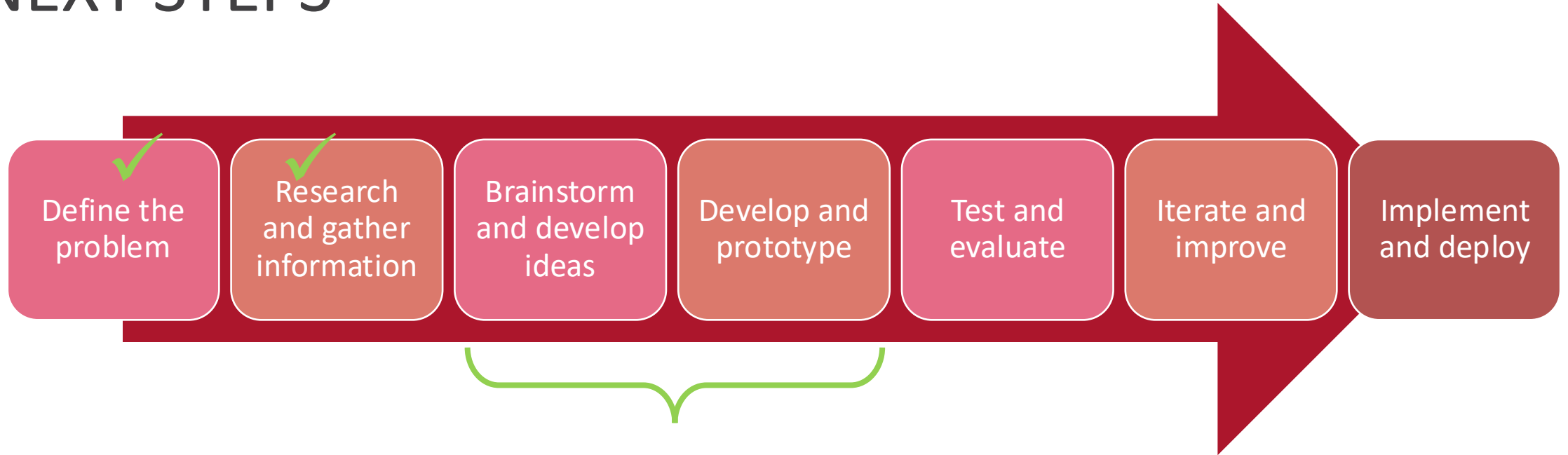
CPT CODING					
CODE	DESCRIPTION	WRVU	FACILITY RVU	ESTIMATED TRVU	MEDICARE CONVERSION
43239	EGD, collection of one of more biopsies	2.39	8.84 (NF)	11.49	\$376.18
43239	EGD, collection of one of more biopsies	2.39	1.37 (F)	4.02	\$131.61
43235	EGD, collection of specimen with brush/wash	2.09	6.59 (NF)	8.93	\$292.37

REGULATORY ASSESSMENT

CLASSIFICATION INFORMATION					
FDA CFR No.	Product Code	Class	Review Panel	Product Code Description	Submission Type
876.5980	FEG OR FDX	II	Gastroenterology/Ur ology	Gastrointestinal tube and accessories OR Endoscope and Accessories	510(k)

REGULATORY STRETEGIES		
Type of Submission:	☒ Traditional 510(k)	☐ Q-Sub/Pre-Submission Meeting
	☐ Special 510(k)	☐ PMA
	☐ Abbreviated 510(k)	☐ PMA Periodic Report
	☐ De Novo 510(k)	☐ PMA Supplement
	☐ Letter to File	☐ 30-Day PMA Notice
	☐ General Wellness	☐ 30-Day PMA Supplement
	☐ None	☐ 135-Day PMA Supplement
	☐ Other (IDE pre-submission)	☐ Real-Time PMA Review

NEXT STEPS



Immediate Next Steps:

- Develop modified nasogastric tubes for Dr. Peterson's IRB
- Design and develop prototypes that optimize mucus collection either via endoscope, nasally, or both.
- Would potentially lead to patent/IP creation.



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